

Internalization of Experience

Cognition from Action

I move, therefore I am.

—HARUKI MURAKAMI¹

In the beginning was the act.

—GOETHE'S *FAUST*

Not hearing is not as good as hearing, hearing is not as good as seeing, seeing is not as good as knowing, knowing is not as good as acting; true learning continues until it is put into action.

—CONFUCIUS ²

Theories of cognition have evolved through multiple stages. The first was the outside-in, empiricist view, postulating that the brain is an associational, representational device that analyzes the world around us and makes judgments. Then came the Pavlovian reflex theory, which did not make much space for cognition; everything was a hierarchy of associative reflexes. Similarly, the behaviorist paradigm argued that there is no need to think about cognition as actions can always be explained as a response to immediate external cues. In response to these views, a hard-thinking minority argued that behavior cannot be understood simply as an input–output function and activity

1. <https://www.goodreads.com/quotes/503630-i-move-therefore-i-am>.

2. Confucius's formulation of the problem is also representational. Although it acknowledges the paramount importance of action, it starts with the senses.

in the hidden layers of the brain is critical (Chapters 1 and 3).³ In this chapter I discuss that *internally organized activity* in these hidden layers, detached from immediate sensory inputs and motor outputs, is a necessary condition for cognition.

DISENGAGEMENT OF THE BRAIN FROM ITS ENVIRONMENT

Humans, and likely other animals, can imagine into the future and recall the past; we spend a great deal of time on such mental activities. I suggest that these activities arise from the brain's ability to break free from external control. The core idea is that cognition depends on prior action-based experiences of the world, which allow internally generated sequences to test "what if" scenarios and anticipate the possible consequences of alternative actions without actually taking them. This process then helps to select future overt actions. To unpack my hypothesis, consider the three possible brain network scenarios shown in Figure 5.1.

In a small nervous system (left panel), the connections between output and input networks are short and simple. This hypothetical small brain's main goal is to predict the consequences of the animal's actions. Such a brain can make experience-based predictions in a simple environment and at a relatively short time scale. These predictions are generated through learning rules that either apply across generations (evolutionary adaptation) or within the lifetime of the organism. As a result of stored past experience, organisms can deal with a future occurrence of similar situations more effectively. This is the fundamental organizing principle of adaptive systems, of which the brain is a prime example. For example, animals learn to predict the occurrence of food or an unpleasant air puff after some training.⁴

3. Tolman's cognitive map theory (1948) was a major departure from the outside-in framework. As opposed to trial-and-error or "instrumental response" solutions, Tolman (1948) speculated that rodents use a mapping strategy (i.e., a "cognitive map") created through exploration to solve maze problems. His thinking fueled John O'Keefe's pioneering physiological experiments (O'Keefe and Dostrovsky, 1971; see Chapter 6).

4. According to the Pavlovian theory, the initially "neutral" conditional stimulus becomes a "substitute" for the unconditioned stimulus after their systematic pairing. Thus, the neutral stimulus will also induce salivation because neurons "representing" the conditional stimulus will become connected to the neurons representing the unconditioned signal. In turn, the latter neurons trigger an inborn stimulus-response reflex (outside-in). One of Pavlov's students, Piotr Stevanovitch Kupalov, was already unhappy with this explanation (Giurgea, 1974) and, along with my mentor, Endre Grastyán, argued that the conditional signal does not become a substitute for the unconditional signal but a new goal, which acquires significance for the animal through active investigation of the conditional signal (Brown and Jenkins, 1968; Grastyán and

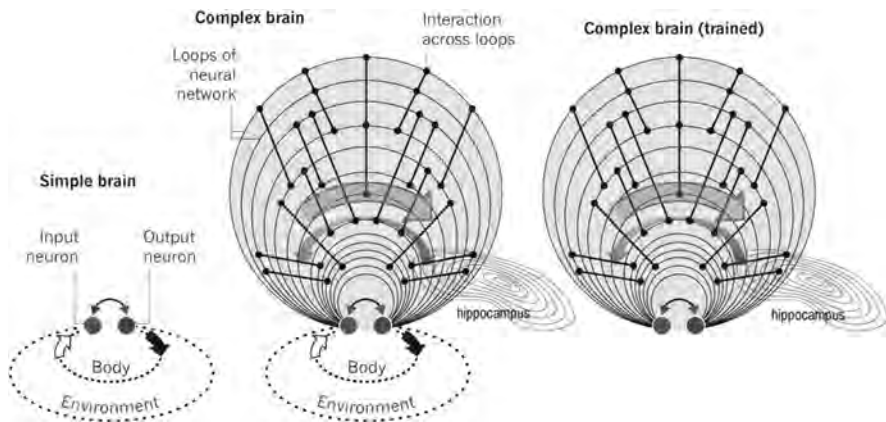


Figure 5.1. Externally driven and self-organized brain operations. A: Small brains contain simple neural networks. The output moves sensors, which scan the surroundings or the body so that the brain can predict the consequences of its action based on prior experience (phylogenetic or ontogenetic) in similar situations. B: In more complex brains, multiple interacting loops of increasing lengths and layers improve prediction of more elaborate events in more complex environments and at longer time scales by creating an efficient memory mechanism (*postdiction*). C: After extensive training, the loops can sustain self-organized, long-lasting neuronal sequences without reliance on external cues and can, therefore, support cognitive operations such as memory, planning, and imagination. Learning and disengagement from the sensors take place in parallel. I show them separately here only to illustrate the principle. Modified from Buzsáki (2013).

More complex brains are organized in a “multiple loop” pattern. These parallel loops, each with increasing levels of complexity of wiring and temporal dynamics, are inserted between the output and input (middle panel). For example, in mammals, the most direct circuit between sensors and muscle activation is the monosynaptic spinal cord/brainstem connection. More complex loops connecting these same inputs and outputs include the thalamocortical system and bidirectional communication between the neocortex and several subcortical structures. In addition, a side loop of the hippocampal system is dedicated to generate sequential neuronal activity such that content in the thalamocortical loops is organized into orderly trajectories, allowing simultaneous postdiction (memory) and prediction (planning) operations (as elaborated in Chapter 13). Large and small brains share the same main goal: to predict the future

Vereczkei, 1974; Kupalov, 1978; for a review, see Buzsáki, 1983). Under this hypothesis, action is essential because it is the motor output that adjusts the sensors to optimally sense the conditional stimulus so that it becomes a meaningful signal.

consequences of their actions. However, the multiple interacting loops and the hippocampal postdiction-prediction system allow more sophisticated brains to predict action outcomes more effectively at much longer time scales and in more complex environments.

Efficient predictive operations in a complex environment require the storage of a vast amount of past experience to compare the current situation to similar occurrences in the past, along with the ability to evaluate and weight the importance of a range of possible actions. Many of these operations require brain computation to continue long after the disappearance of transient sensory inputs and the completion of actions. Such persistent activities are needed to compute the importance of the items perceived and the actions performed. Learning, a result of both phylogenetic and ontogenetic accumulation of experience, makes the longer, more complex neuronal loops “smarter” or more effective by enabling them to interpolate and extrapolate the events encountered by the organism.

Self-organized activity also allows brain networks to disengage from inputs and overt actions (right panel). I propose that, in a trained brain, communication from the action systems to sensory and higher order areas via the expansion of corollary discharge circuits (Chapter 3) allows for such operations.⁵ In other words, the corollary discharge mechanism supplemented with a memory loop can substitute for the feedback provided by the body and environment. This virtual action is not translated to muscle movement or heart rate change, yet it is interpreted by the brain as if those actions could have occurred. Such internalized actions—we can call them simulations of real actions—allow the brain to inspect its own computations; evaluate internally generated, imagined scenarios; and estimate the consequences of potential actions, all without overt inputs or outputs (Figure 5.2).⁶

For a brain network, there is no difference between sensory inputs or activity conveyed by the same upstream neuronal group in the absence of external inputs (see Figure 1.2). Without external constraints, disengaged processing in the brain can create an internalized “virtual world” and new knowledge through vicarious or imagined experience, tested against preexisting and stored knowledge. This process—which most scientists and philosophers would call cognition⁷—provides dramatic advantages in predicting the consequence of

5. Mátyás et al. (2010); Mao et al. (2011).

6. This process is expected to develop during the course of evolution and should be distinguished from the Taoist idea of *wu-wei*, the “nonaction with action” or “unremitting alertness” and from the Buddhist internalized focusing, all of which are practiced during individual lives.

7. Merleau-Ponty (1945/2005), Gibson (1977), and their intellectual heirs developed ideas, similar to those I outlined here, which have come to be known as “embodied cognition” (Thelen, 1989; Gregory, 1980; Maturana and Varela, 1980; Beer, 1990; Brooks, 1991; Varela et al., 1991;

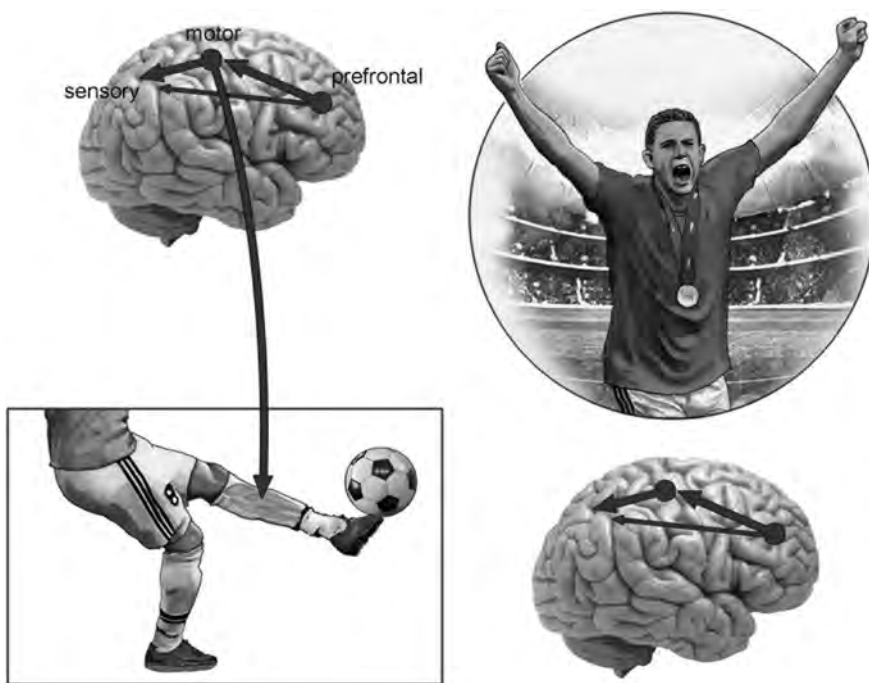


Figure 5.2. The brain can disengage from immediate control of the body. *Left:* Whenever a motor command is sent to the muscles, several other brain regions are informed that an output signal was sent (arrows). Prefrontal neurons activate motor neurons and both areas send a corollary discharge (Chapter 3) to sensory areas. *Right:* After the corollary discharge system is calibrated by experience, involving feedback from the body and environment, activation of the command signal can simulate real actions by activating the same target circuits but without sending signals to the muscles. Such disengagement from immediate muscle control (“internalized or simulated action”) is hypothesized to be the neuronal basis of imagination and planning. A “thought” might be regarded as a buffer for delayed action.

actual behavior in complex environments and at long time scales. Of course, the validity of these newly created ideas should be tested eventually in real-world interactions. Until then, the internal content remains a belief. In short, cognition is time-deferred action.

An important aspect of the framework just presented is that most brain structures have two uses, which I detail in the coming examples. At times, they are connected to sensory input or motor output, directly or indirectly, and vary

Jeannerod, 1994; Clark and Chalmers, 1998; O’Regan and Noë, 2001; Llinás, 2002; Noë, 2004; Buzsáki, 2006; Prinz et al., 2013; Goodrich, 2010). While I share the general philosophy of the situated, embodied mind (i.e., that brains gain their smartness from the body, environment, and other brains), my approach is from a neurocircuit perspective.

their cell assembly contents at the pace of changing sensory inputs. At other times, they rely largely on their internal dynamics, often maintained by brain rhythms (Chapter 6).⁸

I do not claim that processes like input-dependence, self-organized activity, and output disengagement developed sequentially or independently. On the contrary, these mechanisms are strongly intertwined, and the degree to which they dominate in each brain depends on the complexity of the circuits, the availability of external cues, and likely other factors. Thus, the emergence of cognition is gradual and quantitative rather than step-like and qualitative. Let me illustrate the “emergence of cognition” hypothesis by concrete examples. The text will inevitably get a bit technical here and there, but I try to keep it simple.

INTERNALIZATION OF HEAD DIRECTION SENSE

The Aristotelian inheritance in cognitive neuroscience is perhaps most evident when people talk about our five senses. The eyes, ears, nose, skin, and tongue contain sensors for seeing, hearing, smelling, touching, and tasting.⁹ Any sense beyond these classical ones is still referred to as a “sixth sense.” But there are more, including our senses of temperature, pain, balance, motion, and head direction. Of these, the sense of head direction is perhaps the simplest as it has only a single dimension: the direction the head points to.

Neuronal substrates of the head direction sense were discovered by James Ranck at the State University of New York Downstate in 1984.¹⁰ He observed that neurons in the post subiculum, a structure that is part of the hippocampal system, fire robustly when the animal’s head points in a specific direction, and he called them *head direction cells*. For example, when the rat faces a corner of the testing box, one set of neurons become active. When the rat turns its head to right, another set of head direction cells will turn on their activity. For each of the 360 degrees, a different set of neurons fires with equal probability (Figure

8. We have already discussed a striking example of this process, the “internalization” of early spindles, which depends on external stimuli before it becomes a spontaneously occurring sleep spindle.

9. The five senses or “wits” were originally derived from the faculties of the soul, described in the three-book treatise, *De Anima*, by Aristotle. Interestingly, he first lumped taste together with touch before treating it as a separate sense.

10. Jim Ranck debuted his discovery at small meeting that I organized for Endre Grastyán’s sixtieth birthday in Pécs, Hungary, in 1984 (Ranck, 1985). All participants, including John O’Keefe, the discoverer of hippocampal place cells, immediately recognized the importance of Jim’s unexpected finding. For a comprehensive review, see Taube (2007).

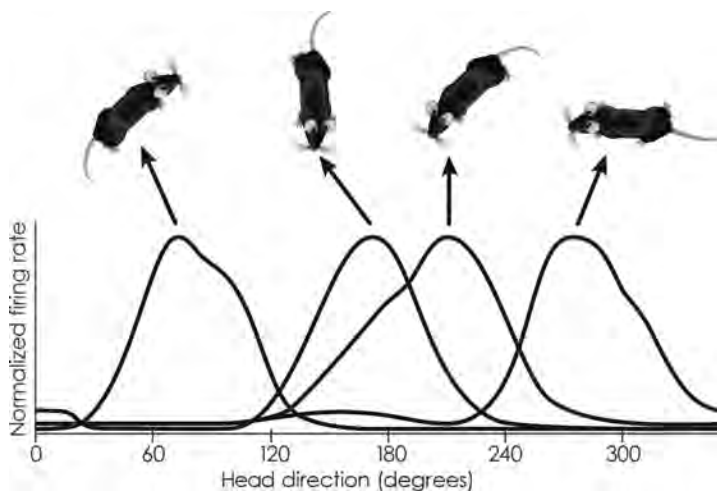


Figure 5.3. For each head direction, a subset of head direction cells become active. Firing patterns of four head direction cells are shown, each firing maximally when the mouse's head faces particular directions.

5.3). The head direction *system* consists of many neurons in a serially connected brain network, including the brainstem, mammillary bodies, anterodorsal thalamic nucleus, post subiculum and entorhinal cortex, lateral septum, and parietal cortex. This widespread representation of head direction in so many brain structures indicates that this sense is of fundamental importance for the brain.

Researchers have tacitly assumed that head direction neurons are largely controlled by peripheral inputs, similar to other sensory systems. Indeed, numerous experiments in multiple laboratories show that head direction neurons are controlled by vestibular, proprioceptive, visual, and ancillary inputs. Such responses, of course, are expected in a sensory system because neurons in all sensory systems respond to the relevant sensory modality. But I often wondered what such “sensory” neurons do when there is no sensation.¹¹ If they are active without their dedicated outside inputs, do they show any organized, cooperative pattern?

To address this question of how head direction neurons work together, we had to wait until the technology matured to allow recording from a sufficiently large population of neurons. We then compared their dynamic interactions in the tiny

11. Computational models have already postulated that head direction neurons with similar preferred directions fire together. According to these models, the temporally engaged group of head direction neurons (i.e., a “hill of activity” or “an activity packet”) moves on a virtual ring as the animal turns its head (Redish et al., 1996; Burak and Fiete, 2012; Knierim and Zhang, 2012).

anterodorsal nucleus of the thalamus during various behaviors.¹² By training an artificial neuron network, we were able to reconstruct the animal's head position from the population interactions of the recorded neurons. When the mouse's head was turning from left to right, this behavior was reliably reflected in a sweep of neuronal firing sequences as each head direction neuron is tuned to a preferred direction. There was excellent agreement between the measured (real) head direction and the (virtual) direction reconstructed from the neuronal firing.

The exciting part of the experiment began when the mouse fell asleep. Because during sleep all peripheral inputs are either absent or constant, one might expect that head direction neurons will be silent or just fire randomly. Yet, to our surprise, not only did head direction neurons continue to be active, but they replicated the sequential activity patterns seen in the waking mouse. Neurons coding for nearby orientation continued to fire together, while neurons with opposite head direction tuning remained anti-correlated. During rapid eye movement (REM) sleep, when brain electrophysiology is strikingly similar to its awake state, the neural patterns moved at the same speed as in the waking animal, recapitulating the activity seen when the mouse moved its head in different directions during waking. In essence, from the temporally shifting activities of cell assemblies, we could determine the mouse's virtual gaze of direction during REM sleep. During non-REM sleep, the activity was still perfectly organized, but now the virtual head direction drifted ten times faster than it did in waking and in REM sleep.¹³ Because no head movement accompanied these changes in the sleeping animal, the temporal organization had to rely on internal mechanisms rather than external stimuli.

The observation that the neuronal code of the head direction sense is preserved during sleep suggests that the system is maintained by a self-organized mechanism.¹⁴ Of course, this self-organized dynamic is expected to remain active in the waking animal and should contribute to some function. For example, sensory inputs can be combined with the internally generated prediction of head direction to rapidly adapt to reconfigured environments. In case of ambiguous or conflicting signals, self-organized mechanisms may generate the brain's "best guess" by interpolating across input signals or extrapolating the correct position of the head direction vector from limited or ambiguous sensory information.

12. Adrien Peyrache was the key player in these experiments (Peyrache et al., 2015).

13. The several-fold faster virtual head direction-related neuronal activity during non-REM is in sync with the previously described faster dynamics of non-REM sleep in cortical networks (Buzsáki, 1989; Wilson and McNaughton, 1994; Nádasdy et al., 1999).

14. "Self-organized" is the term most often used in physics and engineering; it refers to autonomously generated patterns, "changing from unorganized to organized," or any change with decreasing entropy (Winfrey, 1980).

Overall, these experiments illustrate how internally organized mechanisms can improve the brain's interpretation of the external world. Internal computation continues even without external cues and can perform both interpolation and extrapolations (also called *prediction*; Chapter 2). When I close my eyes, my head direction sense persists. But the self-organized head direction system needs to be calibrated by active exploration to exploit its potentials.

FROM PHYSICAL NAVIGATION TO MENTAL NAVIGATION

When you emerge from a subway station in an unfamiliar city with a map in your hands, the first thing you need to do is properly orient the map. North on the map should orient to north in the world. Second, you have to locate yourself—that is, find your subway station. In addition, the vertical and horizontal grid lines on the map (a grid) can then help you to find museums, your hotel, and other places of significance. Third, you need distance calibration to determine whether your target is within walking distance or if you would be better off taking a cab. Now you can use the grid to identify any location or the relationship between two arbitrary locations. The alternative to this map-based navigation would require you to memorize a long list of street names, number of steps, turns to be taken, and so on. This latter strategy works well for moving from one place to a few other places. However, when the goal is to get from anywhere to anywhere else, maps are superior to route descriptions.

To construct a map, someone needs to explore every single locale of the environment. Each location can then be uniquely described by the intersection of horizontal and vertical coordinates on the map. The advantage of this approach is that it allows us to calculate the most efficient route between any two points, thus minimizing distance or time, including potential road blocks or shortcuts. Physical maps are a prime example of the “externalization” of brain function (Chapter 9). Instead of carefully exploring and memorizing the surroundings in each brain, as other animals must do, humans can navigate effectively and without prior exploration as long as they can read a map prepared by someone else.¹⁵ Without speech or other direct communication, maps provide clues about landmarks, distances, and specific locations.

15. The everyday use of maps is rapidly shrinking in modern societies. With the wide availability of global positioning system (GPS), travel and goal finding have reverted back to the locale (list) system. No familiarity with the real world or abstract maps is needed; just follow the verbal commands. While the immediate convenience of GPS-assisted navigation is clear, its exclusive use may deprive people of their spatial sense. Imagine the impact of an attack on or

An entire system in the brain, consisting of several interactive structures, is dedicated to the cognitive map that supports spatial navigation. Head direction neurons provide orientation guidance, the map grid is represented by entorhinal “grid cells,” and self-localization is assisted by hippocampal “place cells.” The source of distance calibration is still debated, but self-motion remains a leading candidate mechanism.¹⁶ The positive impact of practice in navigation on the hippocampal system has been demonstrated by a series of celebrated studies by Eleanor Maguire and colleagues at University College, London. These researchers showed, using structural imaging of the brain, that the posterior (tail) part of the hippocampus in taxi drivers who spent at least three years learning their way around London was larger than in matched controls. Thus, taxi drivers’ brains grow on the job.¹⁷ This is likely the case whenever navigation skills are increased by practice.

Grid Maps in the Entorhinal Cortex

Much like the line grid on a physical map, spatial firing patterns of grid cells provide a metric for the neural representation of space. However, instead of the square grid of typical city maps, grid cells tile the two-dimensional surface of the environment in a honeycomb fashion. This pattern becomes apparent to

failure of GPS navigation satellites. Some people would not even be able to find their way home. For this reason, there is an effort to construct “dead reckoning” devices which do not rely on satellites but track the exact route of travelers (e.g., combat soldiers) and can assist in guiding them back to home base.

16. A significant milestone in our understanding of cognitive representation of space in the brain was the treatise by O’Keefe and Nadel (1978). Although there are hundreds of reviews on the navigational aspects of the hippocampus system, this volume remains the most eye-opening, thought-provoking, and comprehensive treatment of the subject. It has remained the de facto reference for almost everything we have claimed about the hippocampus over the past decades. On October 6, 2014, *The New York Times* heralded that John O’Keefe, Edvard Moser, and May-Britt Moser received the Nobel Prize for Medicine and Physiology for the discovery of a brain GPS that “helps us know where we are, find our way from place to place and store the information for the next time.” The three scientists’ discoveries “have solved a problem that has occupied philosophers and scientists for centuries—how does the brain create a map of the space surrounding us and how can we navigate our way through a complex environment?” Because the neuronal mechanisms of spatial navigation have been detailed in several excellent reviews (O’Keefe and Nadel, 1978; O’Keefe, 1991, 1999; O’Keefe and Burgess, 1996; Burgess and O’Keefe, 2011; Buzsáki and Moser, 2013; Moser et al., 2008, 2014, 2017; Hasselmo and Stern, 2015; Mehta, 2015; Redish, 2016; McNaughton et al., 2006; Connor and Knierim, 2017), I will only briefly summarize the key papers of the large literature on navigation and the hippocampal-entorhinal system here with a bias to help expose my own views.

17. Maguire et al. (2000).

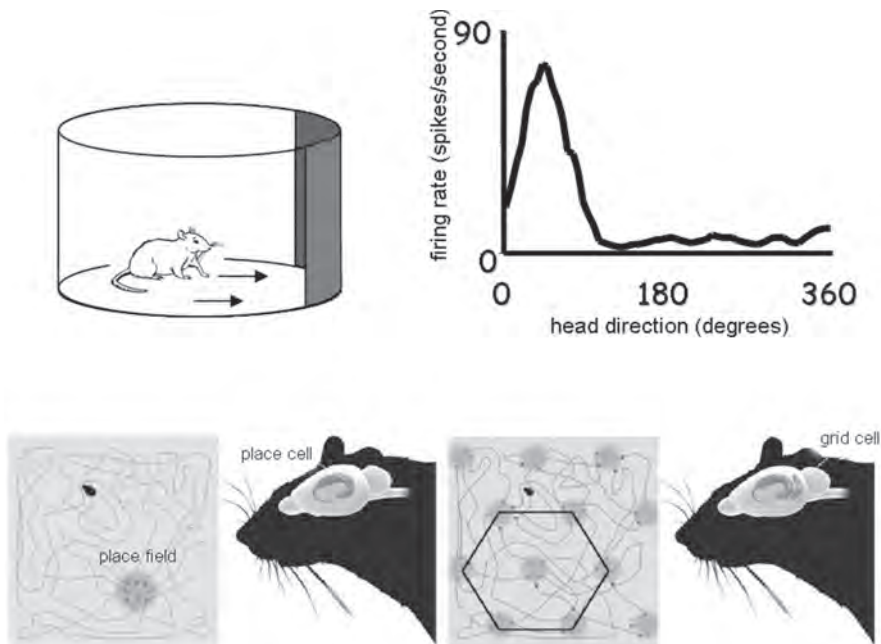


Figure 5.4. Neuronal components of navigation. *Top:* Tuning curve of a head direction neuron from the presubiculum while the rat's head is facing east. *Lower left:* Singular place field of a hippocampal pyramidal neuron during exploration of a rectangular platform. Lines indicate the rat's path; circular area is where the place cell fired most (place field). *Lower right:* Grid cell firing (from the medial entorhinal cortex) while the rat has explored the rectangular platform. Note increased firing rates at the apexes of a hexagon.

the experimenter when the spikes emitted by a single grid cell are projected onto the floor of the experimental apparatus. Think of a TV screen, where a stationary-looking picture is formed by the illumination of many points in fast succession, compared to the slow temporal resolution of the eye. For grid cells, the moving beam is the rat's movements. To see the hexagonal pattern, we have to wait patiently until the animal passes through each pixel of the environment a few times. When enough spikes are emitted during a long exploration, the hexagonal grid firing pattern becomes clear (Figure 5.4).

Neighboring grid cells have a similar pattern but with a slight spatial offset. These cells are arranged as semi-independent modules or clusters along the dorsal-to-ventral axis of the medial entorhinal cortex, where grid scales vary from small to large. This arrangement is like overlaying multiple maps of the same environment with low-, intermediate-, and high-resolution on each other. When the firing patterns of many grid cells are superimposed, they precisely

tile the entire floor of the testing apparatus. Their combined activity amounts to a sequence of population vectors or a trajectory (Chapter 4) that describe the animal's position with high precision at any time because each position evokes a unique firing pattern among grids of different sizes and locations.¹⁸ This self-independent or so-called *allocentric* representation describes the relationship of each position to any other position: a map.

Place Cells and Place Fields

In contrast to the entorhinal grid patterns, hippocampal neurons display highly flexible spatially tuned patterns, called *place fields*. Typically, in a given environment, most pyramidal neurons are silent, and the minority forms a single place field: that is, the neuron becomes active only when the animal reaches a particular location in the testing apparatus. This response is why John O'Keefe named them *place cells*. The size of the place fields is small in the dorsal part of the hippocampus and grows in size toward the ventral pole, in register with the grid size increase of the topographically connected medial entorhinal cortex. Because of the discrete place fields of hippocampal neurons, they carry much more specific information than the entorhinal grid cells. A large population of place cells with unique locations can provide a reliable map and faithfully track the animal's position in the environment.¹⁹ Surprisingly, though, in contrast to the entorhinal grid map, the hippocampal map is not static. If the animal is placed in a different environment, a different set of place cells becomes active. Neighboring place neurons in one environment may have a very different spatial relationship with one another. For example, a pair of place cells representing adjacent pieces of the floor in one apparatus may become silent in another, or just one of them may remain active but at a different position. If both of them continue to fire, the distance between their preferred firing locations may be different in the two mazes. Thus, each environment

18. The discovery of grid cells by the Moser group was a spectacular moment in neuroscience (Hafting et al., 2005). Its beauty has captured the attention of many researchers outside spatial navigation and recruited many new investigators into the field. Initially, it appeared that mapping and navigational functions, attributed to the hippocampus, resided in the entorhinal cortex, freeing up the hippocampus for other functions. The pendulum has swung back and forth a few times over the past decade, intensifying the debate and forcing us to think in a larger context. Despite numerous models and experiments, there is no agreed-upon explanation of how the grid pattern is supported by the circuits in which they are embedded (i.e., the superficial entorhinal layers, the presubiculum, and parasubiculum). It is becoming clearer that functions do not reside in this or that structure but emerge through their interactions.

19. The ability of hippocampal place cell assemblies to track the animal's position in the environment was first demonstrated in a landmark paper by Wilson and McNaughton (1993).

is represented by a unique combination of active place cells and place fields.²⁰ Even seemingly minor changes in the testing apparatus or the variables of the task may strongly alter the firing pattern of the hippocampal place cells.

The spatial layout of individual place cells (i.e., the map) is not related to the location relationships of place cells in the hippocampal circuitry. Two neighboring pyramidal neurons are just as likely to represent adjacent or distant patches of the environment. Instead, place cells dynamically and relatively randomly reconfigure under various conditions. Largely because of this arrangement, the densely connected hippocampal recurrent collaterals can generate discrete maps individualized to the many environments an animal visits in its lifetime.²¹ To envision such relationships, imagine that many place cells on a piece of paper are interconnected with rubber bands. When you crumple the paper, the different place cells will occupy a new set of displacement relationship to each other. These distances on the surface of the wrinkled paper may be thought of as the synaptic strengths between them. Now you can flatten the paper and crumple it again a bit differently; this gives another set of relationship, and so on. Thus, the same set of neurons can give rise to many different constellations of relationships, each of them representing a map.

Of course, a map is just a static positioning tool, a reference frame, in which the spatial relationships among landmarks assist in defining the animal's own location in the environment. This relationship is described eloquently by O'Keefe²²: "Each place cell receives two different inputs, one conveying information about a large number of environmental stimuli or events, and the other from a navigational system which calculates where an animal is in an environment independently of the stimuli impinging on it at that moment. . . . When the animal had located itself in an environment (using environmental stimuli) the hippocampus could calculate subsequent positions in that environment on the basis of how far, and in what direction the animal had moved in the interim. . . . In addition to information about distance traversed, a navigational system would need to know about changes in direction of movement either relative to some environmental landmark or within the animal's

20. The unique firing patterns in different contexts is a segregation mechanism (also often called "orthogonalization"; Marr, 1969), which is believed to be accomplished by interactions between the numerous independently modulated granule cells and the small number of mossy cells whose axons massively project back to the granule cells (Senzai and Buzsáki, 2017). The pattern segregation and integration mechanisms and the assumed role of attractor dynamics in these processes are discussed by McNaughton and Morris (1987), Treves and Rolls (1994), and Knierim and Neunuebel (2016).

21. Samsonovich and McNaughton (1997).

22. Chapter 11; p. 499 in Andersen et al. (2007).

own egocentric space.” However, a map is useful for the animal only after it has already learned the significance of the relationships among the landmarks through active exploration.

As the Crow Flies: Navigation and Self-Motion

Bruce McNaughton and Carol Barnes at the University of Arizona took an important step in linking the static representation of maps to animal navigation through carefully designed experiments and reinterpretation of existing data. Their first key insight was that head direction information is an integral part of the navigation system. The second insight was that self-motion is the primary source of spatial information leading to place-specific firing.²³ In other words, visual landmarks and other stimuli in the environment gain their meaning by action (Chapter 3), although they did not explicitly spell out this link.

Several experiments support the primacy of action in navigation. First, the firing frequency of place cells depends on the animal’s speed. Second, on a running track, a robust place field observed on the left-to-right journey is often absent on the return run. That is, place fields are not absolute descriptors of the x - y coordinates of the environment; they also depend on context, such as which direction the rat is facing and how head direction orients the entorhinal-hippocampal map. Third, under certain circumstances, the same physical space can be represented by different sets of place cells depending on other contextual variables. For example, when a rat is placed into an otherwise familiar environment in total darkness, new place fields can emerge and persist in subsequent light. Thus, place fields can arise independently of the visual landmarks, although landmarks can control the expression of place cells. Fourth, changing the features of a landmark without changing its position can affect the map. For example, after a white cue card in a symmetric apparatus (cylinder) is replaced with a black one, the place cells stay put relative to the radial axis of the cylinder. However, after several trials in which the white and black cards are alternated, the place-field distributions of several neurons become largely uncorrelated across the two conditions as the animal learns to distinguish the

23. McNaughton and his long-term collaborator Carol Barnes were postdoctoral fellows in O’Keefe’s lab. They provided firm experimental evidence and drew attention to the importance of local cues and body-derived signals in self-localization in navigation (McNaughton et al., 1996). On computational arguments, Redish and Touretzky (1997) also emphasized the need for a path integrator independent of the cognitive map. See also Gallistel (1990).

two cues.²⁴ Fifth, when a rat is repeatedly disoriented (by turning it by hand multiple times before placing it into a cylinder with few landmarks), its place fields tend to destabilize. Sixth, when the distance between the start and goal boxes changes, only a fraction of neurons remain under the control of distant room cues. Many of them realign as a function of the distance from either the start box or goal box.²⁵

The list goes on. The profound implications of these observations led McNaughton, Carol Barnes, and their colleagues to postulate a hypothetical “path integrator” system, a sort of a body cues feedback-supported guiding mechanism, based on information from local visual and somatosensory cues, proprioceptive feedback from the body (e.g., muscles, tendons), the number of steps taken, vestibular input (translational and rotational head accelerations), and, likely, corollary discharge from self-motion activity. Integrating self-motion allows animals to move through space even without prominent landmarks while keeping track of their starting location. The path integrator system can operate without a prior spatial reference or even in complete darkness by calculating the distance traveled and the turns the animal made.

Dead reckoning, another name for path integration, was one of the earliest forms of marine navigation. From a home base, the captain calculates the movement of the ship based on heading, speed, and time. Multiplying speed by time provides the distance traveled. The sequence of distances and heading direction changes is noted on a chart. This route chart can always be used to calculate the straight path to return home. Dead reckoning was the main navigation strategy for sailors in the Mediterranean Sea for centuries before Portuguese sailors introduced maps and landmark (more precisely “skymark” or celestial) navigation.²⁶

24. Several experiments convincingly demonstrate that distant visual cues can have a large impact on place cells. For example, when the walls of the behavioral apparatus are rearranged, the size and shape of place fields often follow such modifications by enlargement, elongation, or even splitting the place field into two (Muller and Kubie, 1987; Kubie et al., 1990; O’Keefe and Burgess, 1996; Wills et al., 2005; Leutgeb et al., 2005).

25. Redish et al., 2000; Gothard et al., 2001; Diba and Buzsáki, 2008).

26. Christopher Columbus used dead reckoning to discover the New World and safely return home. The term “path integration” (*Wegintegration*) was introduced by Mittelstaedt and Mittelstaedt (1980) to reflect the idea that a continually updated representation of direction and distance from a starting point is a reflection of signals generated by self-motion (also known as *idiothetic*) as opposed to the intermittency of navigation via landmarks, which may not be sensed continuously. These authors also wrote that the updating of position is much more accurate during active motion as opposed to passive shifts. This is familiar to us: the passenger’s sense of the surroundings is always poorer than that of the driver. Before the work of the Mittelstaedts on gerbils, Wehner and Srinivasan (1981) reported that desert ants can return directly to the nest after an outward journey of hundreds of meters, using the shortest path. See also Whishaw et al. (2001).

Navigation by path integration is intuitively appealing.²⁷ Not only sailors but also animals can find their way without landmark cues. A gerbil mom can reliably retrieve her pups from within a circular arena and return to her nest at the arena border in complete darkness using an internal sense of direction. When the platform is slowly rotated so that the mom does not notice it, she carries the pups to the original spatial position even though the nest is no longer there. Yet path integration presents some challenges. Every change in speed and direction contributes to error accumulation. As each estimate of position is relative to the previous one, these errors quickly accumulate.

The exact brain networks involved in path integration have been debated from its conception. Initially, the hippocampus was the suspect, based on both recording and lesion experiments. With the discovery of grid cells, the entorhinal cortex became the favorite structure. Although the majority of grid cells are present in layer II of the dorsomedial entorhinal cortex, many neurons in deeper layers respond to head direction, and a subset of these neurons also show typical grid responses.²⁸ The running speed of the animal correlates with the firing rate of some place cells and most inhibitory interneurons. Thus, the conjunction of positional, directional, and translational information in the same entorhinal groups of neurons may enable the updating of grid coordinates during self-motion-based navigation—as the argument goes. However, such correlation alone may not be sufficient because entorhinal neurons receive information about head direction and speed from multiple sources. In support of this idea, damaging the anterodorsal nucleus of the thalamus, a main hub of the head direction system, leads to the disappearance of grid dynamics. Furthermore, position, speed, and head direction information, and even distance, are present in hippocampal and the adjacent subicular neurons as well. Neurons in the parietal and retrosplenial cortex may also be involved in computing the animal's motion relative to environmental and body cues, suggesting that path integration involves multiple systems.²⁹

27. A blind child who is walked from a base to three objects (landmarks) in a large room can immediately find the routes between the three objects on her own, even though she has never experienced those paths. To commute between any two targets, the child must encode the three training paths by dead reckoning, store these paths, and combine two correct stored paths to deduce a new one, a demonstration that a cognitive map can be constructed solely on the basis of dead reckoning navigation (Landau et al., 1984; Wang and Spelke, 2000).

28. The context-invariant firing properties of grid cells led to the suggestion that they are part of a path integration spatial representation system (Hafting et al., 2005; McNaughton et al., 2006).

29. Alyan and McNaughton (1999); Cooper and Mizumori (1999); Leutgeb et al. (2000); Whishaw et al. (2001); Etienne and Jeffery (2004); Parron and Save (2004); McNaughton et al. (2006); Sargolini et al. (2006); Samsonovich and McNaughton (1997); Winter et al. (2015);

A navigation system is only as useful as its ability to navigate the body. How does the abstract hippocampal cognitive map instruct the body to choose the right path in this map? The most dense projection of the hippocampus converges on a subcortical structure called *lateral septum*, which in turn projects to the hypothalamus, the motivational mover of the brain. Indeed, spatial information about the animal's whereabouts is also present in neuronal assemblies of lateral septal neurons.³⁰ Therefore, the lateral septum is an interpreter and translator of the abstract map to motivated action.

Allocentric map representation and self-referenced or *egocentric* path-integration route information work together. The environmental conditions determine which strategy dominates. In cue-rich environments, representations can be updated frequently by changes in the configuration of sensory inputs. In environments with few stationary landmarks or in complete darkness, path integration is the default mode. Do these map and path integration mechanisms need different neuronal networks, or are they different manifestations of the same or overlapping neuronal substrates? We can only guess that navigation effectively exploits both systems. When the goal is to generate a universal solution (getting from anywhere to anywhere else), maps are superior to route descriptions provided that landmarks are available. Planning a route within a map requires creating a mental image of the self and its position on an *allocentric* map. We can gain some insight into how this may happen by looking at how autonomous robots learn to navigate.

Navigating Robots

Like animals, navigating robots also face the problem of locating themselves in the environment. The breakthrough in autonomous robot navigation came from the recognition that perfecting the robot's sensors does not help much with navigation. Sensors are useless unless the robot can actively move them while keeping track of the motion-induced sensory feedback from the environment. Once mapping and robot localization were combined into a single estimation problem, progress accelerated. This insight is nicely expressed

Acharya et al. (2016). People can still learn simple and distinct routes after damage to the hippocampus. However, the hippocampus is indispensable when delays are inserted between the legs of a journey or when novel routes between separate path representations need to be calculated (Kim et al., 2013).

30. Tingley and Buzsáki (2017).

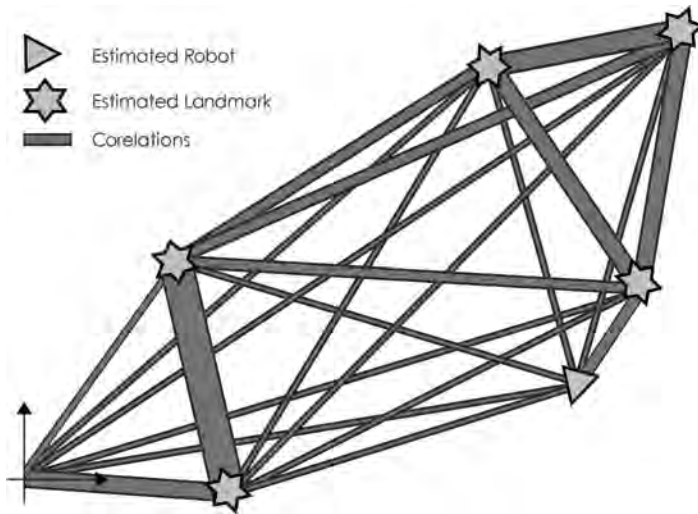


Figure 5.5. Spring network analogy. Springs connect the landmarks and describe their correlations. As the vehicle moves back and forth through the environment, spring stiffness or correlations increase (links become thicker). As landmarks are observed and estimated locations are corrected, these changes are propagated through the spring network. These continuously updated correlations are learned or ‘internalized’ by the robot. Such correlations can be effectively constructed in recurrent networks, such as the hippocampal CA2/3 systems.

Reprinted with permission from Durrant-Whyte and Bailey (2006).

by the acronym for the solution: simultaneous localization and mapping (SLAM), a combination of motor-driven path integration and sensor-dependent landmark detection (Figure 5.5). The movement trajectory of the mobile robot and the locations of all landmarks are continuously estimated and updated without the need for any a priori knowledge. In the process, the robot can construct a map of an environment by fusing observations from different locations while using the map to deduce its location. The estimation accuracy of the robot’s current position is bounded only by the quality of the actively constructed map.³¹ The more the robot moves around, the better the estimation of the relationships among the landmarks becomes. Thus, performance improves with exploration. At the conceptual level, autonomous robot navigation is now considered a solved problem, although numerous practical

31. The *spring map model* is reminiscent of the “cognitive graph” idea of Muller et al. (1996), in which the springs are replaced by the synaptic strengths between hippocampal CA3 neurons.

issues remain in realizing more general solutions, increasing speed and accuracy, and generalizing to multiple environments.³²

This spectacular progress in autonomous robot navigation may soon make them part of our everyday lives. A variety of mobile robots have been tested indoors and in outdoor environments where not all landmarks are fixed. Robotic cars must constantly update their position relative to other moving objects. Recent mobile robots can even (relatively) autonomously navigate through busy urban city centers and interact with pedestrians and pets.³³ These technical developments have gone a long way from Grey Walter's *Machina speculatrix* (Chapter 1), but fundamental ideas about ambulation-induced knowledge collection have remained the same.

If machines with a few microprocessors or insects with tiny brains can solve the navigation problem, why do mammals need a complicated navigation system with millions of interactive neurons? To be fair, autonomous robots are not (yet) perfect, and unmanned cars and trucks strongly depend on global positioning system (GPS; i.e., human made) assistance. Most insects do their business in relatively constant environments so they do not need to construct many independent maps. Their seemingly astonishing navigation abilities depend on rigid computational rules. Ants and bees can proceed directly from one familiar food goal to another by selecting the appropriate long-term vectors and combining them with one another.³⁴ However, they consistently fail in situations that require explicit comparison between separate maps. The mammalian navigation system not only represents more environmental details,

32. Solutions to the probabilistic SLAM problem involve finding an appropriate representation for both the observation model (landmark navigation) and the motion model (dead reckoning navigation) that allows efficient computation of the prior and posterior distributions (Durrant-Whyte and Bailey, 2006). The graph-based SLAM is modeled as a sparse graph, where nodes represent the landmarks and each instant pose state, and edges correspond to either a motion or a measurement event. The advantage of graphical SLAM methods is their ability to scale the environment, so they can generalize from small to related larger environments (see also Brooks, 1991). Many novel nonlinear control systems have been developed recently that have found practical applications ranging from digital "fly-by-wire" flight control systems for aircraft, to "drive-by-wire" automobiles, to advanced robotic and space systems (Slotine and Li, 1991).

33. Unmanned navigation with robotic cars has been studied intensively. These efforts have been largely promoted and financed by the Defense Advanced Research Projects Agency (DARPA; Seetharaman et al., 2006; Urmson et al., 2008; Rauskolb et al., 2008), Google (Google, Inc., 2012), and the European Land Robot Trial (ELROB). It is hard to keep up with the progress in this fast-moving field (Schneider and Wildermuth, 2011; Adouane, 2016).

34. The spatial cognitive abilities of invertebrates, rodents, and humans are reviewed by Wehner and Menzel (1990). For an opposing view, see Gould (1986).

but it can flexibly combine and compare any of those details to symbolize and categorize events, objects, and living things.³⁵

INTERNALIZING ROUTES AND MAPS FOR MENTAL NAVIGATION

To investigate how such combinatorial abilities may have emerged in mammals, we must recall the *dictum* “structure constrains function,” which is especially true in the hippocampal system. Nothing about its architecture and connectivity suggests that it is evolved for spatial navigation but nothing else. The expansively parallel granule cell interface and the strongly recurrent CA2/3 excitatory system, two very different architectural designs, appear ideal for segregating and integrating places and objects. However, segregation and integration are fundamental operations that could benefit numerous brain processes.³⁶ Thus, the hippocampal system, evolved initially for spatial navigation, may be repurposed for other tasks as well.

One Circuit, Many (Apparent) Functions

The hippocampus-entorhinal system has a topographically organized bidirectional communication with the large neocortex. In a rodent, a major fraction of the neocortex deals with controlling motor outputs and processing sensory inputs. In primates, in contrast, a large part is dedicated to computing more complex functions.³⁷ The nonsensory representation in the hippocampus has progressively increased with the enlargement of the neocortex during mammalian evolution (Figure 5.6). Therefore, computation of spatial information might be only a part-time job for the hippocampus in animals with

35. Etchamendy et al. (2003) showed that mice with hippocampal damage cannot flexibly compare previously learned relationships between the arms of a maze and reward, whereas intact mice can.

36. Similar is different! Whether something is deemed similar or different can be judged from a particular point of view (a “classifier”) by an observer. Without an a priori goal, no meaningful classification is possible, or the classification process may turn up numerous minute differences whose meaning cannot be interpreted without a goal. The hippocampus is such a classifier, and the meaning of its computed results (i.e., the segregated items returned to its senders) is interpreted by the neocortex.

37. Because neuroscience’s philosophy followed the footsteps of British empiricism (Chapter 1), it is not surprising that these nonallied, higher order cortical areas are still referred to as “association” cortex.

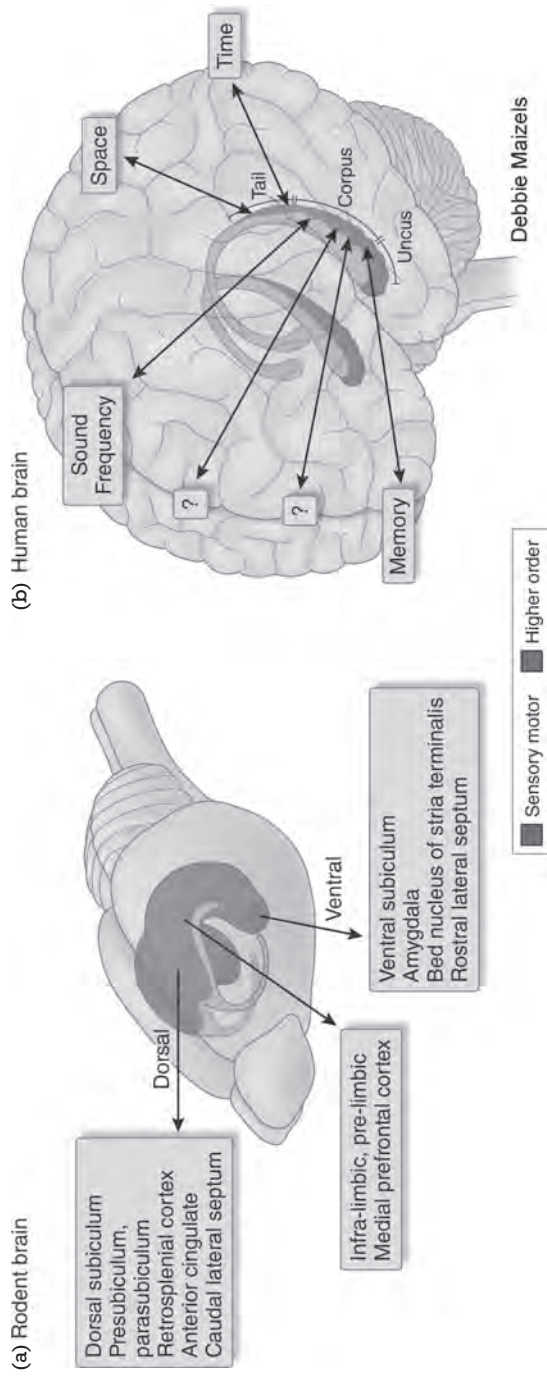


Figure 5.6. Homologous regions of the hippocampus in the human and rat brains. The ventral quadrant of the rodent hippocampus became disproportionately enlarged in primates to keep up with the increasingly larger share of higher order neocortex. Only the relatively small tail part of the primate hippocampus communicates with visuospatial areas. This tail is homologous to the rodent dorsal-intermediate hippocampus. The different connections to and from the segments of the septotemporal axis are shown. Most recordings and manipulations in the rodent brain have been done in the dorsal hippocampus. Adapted from Royer et al. (2010).

larger brains.³⁸ This idea suggests that whatever information the hippocampus receives from whichever parts of the neocortex, the same general computation will be performed. In other words, the hippocampus is “blind” to the modality and nature of the inputs. It processes the messages from the neocortex in the same way irrespective of their origin and returns its judgment to the source.³⁹ Whether it appears to process spatial, temporal, or other information is determined largely by the experimental setup.

If the hippocampus served only navigation, it would remain a mystery how I can jot down these lines without any help from environmental cues in Narita Airport, Japan, where I am currently waiting for my connecting flight. These very same structures are also responsible for seemingly separate functions, such as memory, planning, and imagination all of which must be supported by self-generated cell assembly sequences rather than rely on external or body-derived stimuli. As I explain later, these cognitive functions can be thought of as mental navigation that emerged from mechanisms which were initially introduced for spatial navigation in the physical environment.

38. The septal (dorsal) and midtemporal thirds of the rodent hippocampus receive visuospatial and other sensory inputs from the dorsal entorhinal cortex, whereas the temporal pole communicates mainly with hypothalamic, amygdalar, and prefrontal areas (Risold and Swanson, 1996; Amaral and Lavenex, 2007). In the primate brain, the ventral pole becomes disproportionally enlarged to keep up with the large fraction of the nonallied neocortex and forms the uncus and body. Only the relatively small tail part of the primate hippocampus communicates with visuospatial areas. This tail is homologous to the rodent dorsal hippocampus, and most recordings and manipulations in the rodent brain have been done here. The relationship between the size of the hippocampus and neocortex is unusual in cetaceans. In dolphins and several other aquatic mammals, the hippocampus is embarrassingly smaller than would be expected from its enlarged neocortex partner. Another paradox is that these mammals have little or no postnatal neurogenesis of granule cells (Patzke et al., 2015). One argument is that landmark (or “sea-mark”) navigation is not very useful in the vast waters of the ocean. All you see most of the time is water anyway, so there is no need for a cognitive map. Thus, the hippocampus “shrinks.” In marine parks, the dolphin pools are separated by an underwater net, which rises above the water only a foot or so. I have long wondered why they never jump over the net. Are they not curious enough to investigate what is over there? Or do they have a different concept of space than terrestrial mammals (Gregg, 2013)?

39. For example, subgroups of ventral CA1 pyramidal cells project relatively selectively to the prefrontal cortex, amygdala, and nucleus accumbens. These subgroups become activated contingent on the nature of the task (e.g., anxiety or goal-directed behavior; Cioocchi et al., 2015). However, the hippocampus runs the same “algorithm” in each of these behaviors. See Aronov et al. (2017); Terada et al. (2017).

Forms of Mental Navigation

Starting with the famous patient HM, who had both hippocampi surgically removed in an attempt to alleviate his intractable epilepsy, a consensus emerged that the hippocampus and associated structures are responsible for making memories.⁴⁰ How are memories related to spatial navigation, another postulated hippocampal function? John O'Keefe, the preeminent promoter of the cognitive map theory, speculated that "addition of a temporal component to the basic spatial map in the human provides the basis for an episodic memory system."⁴¹ On the one hand, this suggestion implicitly implies that the hippocampus in rodents and in humans does different things and that rodents lack a "temporal component."⁴² On the other hand, it is not at all clear how adding time to a cognitive map, an allocentric construct that depends on external drive, can provide a mechanism for the most egocentric thing we possess, our collection of episodic memories.

Perhaps if the navigation system can disengage from environmental or body cues, its internalized performance could support memories. Analogous to spatial navigation, two forms of hippocampus system-dependent memories can be distinguished. These are personal experiences (episodic memory or memory for specific instances) and memorized facts (semantic memory or memory for statistical regularities). We are consciously aware of both types and can verbally declare them, so together they are called *declarative memories*. Terms such as "episodic-like" or "flexible" are used to refer to analogous memories in non-human animals.⁴³

40. The amnesic syndrome following bilateral hippocampal lesion was presented by Scoville and Milner (1957). Independent of this celebrated study, Nielsen (1958) concluded after studying a large number of cases that different brain systems are involved in different memories. One system supports the life experiences of the individual and involves a time component (*temporal amnesia*). The other type of memory corresponds to acquired knowledge, facts that are not personal (*categorical amnesia*). See also Milner et al. (1998); Corkin (2013). Several comprehensive reviews have been written on this topic (Squire, 1992a, 1992b; Cohen and Eichenbaum, 1993; Milner et al., 1998; Eichenbaum, 2000).

41. O'Keefe (1999).

42. This argument is fueled by Tulving's (1983, 2002) repeated assertion that only humans have episodic memories because only humans have the ability to "feel" subjective time. His famous patient KC could not recollect any personally experienced events, circumstances, or happenings. KC could not perform mental time travel but could use a map. His semantic knowledge base was relatively normal, and he knew many facts about himself.

43. "Semantic" is from the Greek word *semantikos*, meaning "significant." "Episodes" are combination of things, from the Greek *episodesion*, meaning "linking songs" or "odes" in Greek tragedies. Generalization across episodes means to leave behind the concrete in order to abstract the common links. The distinction between episodic and semantic memory was made by

Episodic Memory

Personal memories are important episodes of our lives that give rise to the feeling of the self and are the sole source of individuality.⁴⁴ To re-experience such first-person (egocentric) context-dependent episodes, we need to project ourselves back in time and space. Endel Tulving of the University of Toronto, Canada, coined the term “mental time travel” for this hypothetical brain operation. Mental time travel allows us to be aware of the past and the future. Going backward in time is known as *episodic recall*, while traveling into the imagined future can be called *planning*. Some people might protest, of course, that the past and future are vastly different, and this distinction should also apply to memory and planning. Indeed, these categories are typically dealt with in different chapters of neuroscience handbooks and assigned to different brain structures and mechanisms. But the brain may not see things this way (Chapter 10).

My first proposal to address these problems is that the brain mechanisms that evolved initially for navigation in physical space by dead reckoning are basically the same as those used for navigation in “cognitive space” to create and recall episodic memory. My second, related proposal is that the neural algorithms evolved to support map-based navigation are largely the same as those needed to create, store, and remember semantic knowledge. My third proposal is that generation of semantic (allocentric) knowledge requires prior self-referenced episodic experience, akin to map creation by dead-reckoning exploration (Figure 5.7).⁴⁵

Tulving (1972) on theoretical grounds and has been supported by numerous lesion, physiological, and imaging data (Squire, 1992a).

44. Becoming an “in-dividual” means acquiring the single sense that my self cannot be divided into parts or subsystems.

45. Although today it is well accepted that the hippocampal system is essential for memory (Milner et al., 1998) and spatial navigation, views regarding the fundamental role of the hippocampus were diverse not so long ago. Here is a quote from the leader of the cognitive map theory, John O’Keefe (1999) “I reiterate the basic tenet of the cognitive map theory that the processing and storage of spatial information is the primary and perhaps exclusive role of the hippocampus in the rat, and that the data that appear to contradict this have been misinterpreted.” Others also postulated that the hippocampus supports different functions in rodents and humans (e.g., Tulving and Schacter, 1990). The idea that the evolutionary roots of episodic and semantic memory systems are the dead reckoning and landmark-based forms of navigation, respectively, are discussed in Whishaw and Brooks (1999); Buzsáki (2005); Buzsáki and Moser (2013). The relationship among place cells and grid cells is not confined to the spatial domain and physical navigation but may represent a common organizational structure of all two-dimensional spaces that are needed for various cognitive computation such as memory, inference, generalization, imagination, and learning (Constantinescu et al., 2016).

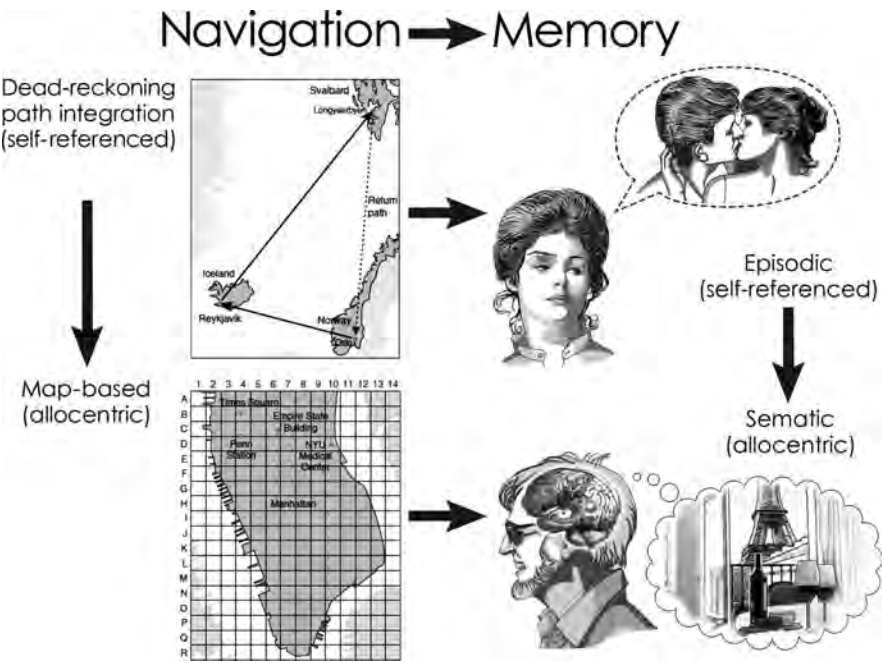


Figure 5.7. Relationship between navigation and memory. *Left:* Path integration (dead reckoning) is based on self-referenced information: keeping track of travel distances (elapsed time multiplied by speed) and the direction of turns. Calculating positions relative to the start location allows the traveler to return to the start along the shortest (homing) path (dotted arrow). *Bottom left:* Map-based navigation is supported by the relationships among landmarks. A map is constructed by exploration (path integration, as indicated by the large arrow from dead-reckoning to map). *Top right:* Episodic memory is “mental travel” in time and space referenced to the self. *Bottom right:* Semantic memory is the explicit representation of living things, objects, places, and events without temporal or contextual embedding. Semantic knowledge can be acquired through multiple episodes with common elements (as indicated by the arrow from episodic to semantic).
From Buzsaki and Moser (2014).

Semantic Memory

In contrast to the observer-dependence of egocentric episodic memories, semantic knowledge is observer-independent.⁴⁶ Semantic memory defines objects, facts, and events of the surrounding world independently of temporal

46. Philosophers usually refer to observer-independent things as the *objective reality*, referring to things in the physical world. In addition, many other human-created ideas also fall into the semantic category, such as names, events, concepts (such as space and time), and imagined things.

context, similar to how place cells and grid cells explicitly define position coordinates in a map. The deep relationship between physical and mental navigation is also illustrated by the practice of Greek orators. They memorized their speeches by imagining that they walked through different rooms of their house, with each room containing a particular topic. Professional jazz musicians also imagine a landscape of notes and navigate through a preplanned path in a concert.

Unlike the one-trial acquisition of episodic memory of a unique event, *semantic information* is typically learned after repeated encounters with the same thing or event. Initially encoded in episodic memory, information gradually loses its contextual features to become generalized and explicit, reminiscent of the place fields, which are omnidirectional; that is, a place cell fires in its place field irrespective of from which direction the animal approaches it.⁴⁷ For example, if you are lucky and discover something, that episode remains precious for the rest of your life. However, if my laboratory and others confirm your discovery by reaching the same conclusion from different angles, it becomes a fact: explicit knowledge understood by everyone in the same general way, irrespective of the conditions of the discovery and confirmation.

Not everyone agrees that episodic experience is necessary to encode semantic information. Indeed, in humans, speech and other externalized functions of the brain allow rapid acquisition of semantic knowledge without the need for personal experiences (Chapter 9). A caregiver can provide the names of objects, which the child then learns. Patients born without a hippocampus can acquire facts and use them efficiently to communicate. However, even though social interaction can provide shortcuts to knowledge, this ability does not undermine the evolutionary origin of episodic and semantic memory from dead-reckoning and map-based navigation, respectively.

47. For further discussion of the relationship between episodic and semantic memories, see Tulving (1972); McClelland et al. (1995); Nadel and Moscovitch (1997); Eichenbaum et al. (1999); Manns et al. (2003); Hasselmo (2012). Place cells explicitly define the coordinates of a map. Similarly, neurons in the hippocampal and entorhinal cortex in epileptic patients implanted with depth electrodes respond selectively to pictures of objects or persons, largely independently of their physical characteristics, and even to their names (Heit et al., 1988; Quiroz Quiroga et al., 2005), which are the defining criteria of explicit knowledge. Experiments on amnesic patients suggest that acquisition of semantic knowledge is possible when brain regions required for episodic memories are damaged. Patient KC was unable to recollect any personal experience, yet he could learn new facts and relations (Hayman et al., 1993; see also Schacter et al., 1984; Shimamura and Squire, 1987). However, his learning was very slow, likely because of the lack of the boosting effect of the episodic system. Three young adults who became amnesic at an early age acquired nearly normal semantic knowledge of world facts and relations despite a severe episodic memory deficit (Vargha-Khadem et al., 1997).

Do Memories Migrate?

The relationships among events and objects, known as *semantic proximity*, shares many features with distance relationships in landmark navigation. Models of semantic relatedness use a metric based on topological similarity, just like the spatial relations among landmarks in the cognitive map.⁴⁸ There is a general agreement that declarative memories depend on the entorhinal cortex–hippocampal system, although debate persists about the fate of consolidated semantic information. One controversy is whether the neural circuits involved in creating semantic and episodic memories overlap or completely separate.

Another controversy is whether memories remain in the circuits where they were created or move gradually from the hippocampal system to the neocortex over time, so that retrieval of consolidated memories no longer requires the structures that originally created them. The latter idea likely derives from the initial description of the cognitive symptoms of patient HM. According to early analyses, he could not learn or remember new declarative memories but could recall virtually anything that happened before his brain surgery. However, subsequent work on HM and similar patients revealed that, after bilateral hippocampal damage, they cannot narrate and lack the ability to “travel mentally” either to the past or the future. Their imaginative and planning skills are as strongly impaired as their ability to place themselves into a spatiotemporal context. Instead, they use their intact logic and semantic knowledge to invent scenarios that might sound reasonable.⁴⁹ For example, if you asked about the house where the amnesic patient grew up, you might be impressed with the recalled details and perhaps events. However, the patient remains unable to narrate a single personal episode, such as a dinner with the family or a fight with siblings, in the proper sequence. In other words, these patients cannot recapitulate or reimagine a situation by embedding themselves into it.

A more recent view, known as the “two-trace” model, suggests instead that although episodic memories are transferred to the neocortex, a copy of

48. Trope and Liberman (2010). The spatial origin of our semantic classification of explicit knowledge is also reflected by the vector distances used for defining relationships among words (Talmy, 1985; Navigli and Lapata, 2010). Such distance representations are also used in “mind maps” and “concept maps.” See <https://wordnet.princeton.edu/>.

49. Such confabulation is a core symptom of Korsakoff’s syndrome, originally described in people with chronic alcohol abuse. Degeneration of the mammillary bodies, due not only to alcohol but also to vitamin B (thiamine) deficiency, drug abuse, or head trauma affecting the diencephalon, as well as to bilateral damage to the temporal lobe, can result in an inability to faithfully recall one’s past (Bayley and Squire, 2002; Downes et al., 2002; Squire et al., 2004; Gilboa et al., 2006). The memory troubles of the character Leonard Shelby in Christopher Nolan’s movie *Memento* (2000) pretty accurately illustrate the symptoms of retrograde amnesia.

the memory is retained in the hippocampus. In support of this view, functional magnetic resonance imaging (fMRI) experiments in human research participants consistently show hippocampal activation during memory recall even for events acquired decades earlier.⁵⁰ An alternative explanation is that the memory trace itself no longer resides in the hippocampus, but that hippocampal networks help to coordinate the sequential recall of items stored in the neocortex, thus providing an egocentric perspective for the items and events of the episode (see Chapter 13).

Overall, comparison of spatial and mental navigation indicates how disengagement of navigation networks from their external dependence could transform them into internalized memory operations.⁵¹ This recognition simplifies our understanding of neuronal mechanisms of apparently different functions. While spatial navigation, memory, planning and imagination are distinct terms, their neuronal substrates and neurophysiological mechanisms are identical or at least similar. Viewed from the brain's perspective, these distant chapters of behavioral neuroscience and psychology can be unified.⁵²

INTERNALIZING ACTIONS: THE “MIRROR” NEURON SYSTEM

Another spectacular example of the disengagement of the brain from its sensory input and motor reafferentation dependence is the mirror neuron system. Like neurons in the hippocampal-entorhinal system that are active during both encoding and recall of experience, many neurons in the neocortex become active not only when we act intentionally but also when we interpret other people's or even our pet's intentional actions. They serve dual functions.

50. Support for the transfer of memory trace can be found in Squire and Alvarez (1995); McClelland et al. (1995); Eichenbaum (2000); Frankland and Bontempi (2005). Many experiments supporting the two-trace model are reviewed in Nadel and Moscovitch (1997).

51. A striking example of how externalized spatial memory may be the precursor to internal memory is the demonstration that the brainless slime mold *Physarum polycephalum* can avoid areas it has previously explored. Drawn into a trap, it can avoid it as if it has constructed a map of the environment (Reid et al., 2012).

52. Functions are often repurposed as conditions change in evolution. The transformation of physical to mental navigation is only one striking and well-researched example. Neuronal mechanisms, such as the grid cell dynamic that supports allocentric (map-like) navigation might also underlie more complex functions, such as logical inference and reasoning (Constantinescu et al., 2016). Grid cell mechanisms could support periodic encoding in multiple cortical areas for abstract categorization and general knowledge organization.

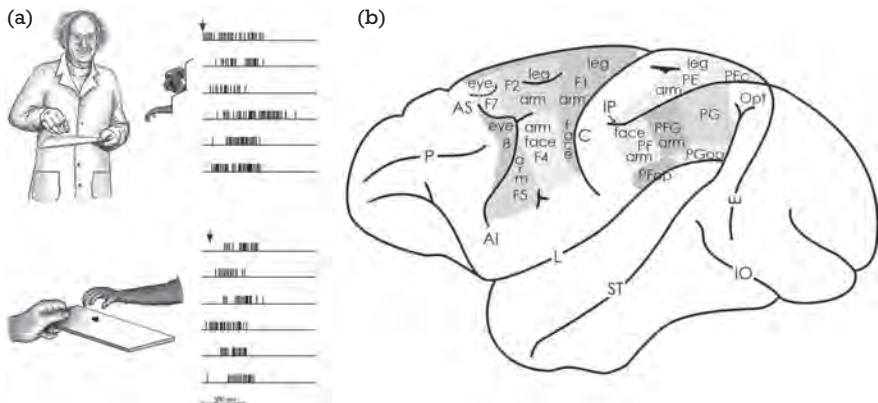


Figure 5.8. Mirror neurons recorded from the prefrontal area of the monkey. A: The neuron is equally active when the monkey reaches for the raisin (*bottom*) or when observes the experimenter reaching for raisin (*top*). Vertical ticks, action potentials of the neuron. B: Shaded areas indicate brain regions from which mirror neurons were recorded.

The serendipitous discovery of these “mirror neurons” is arguably one of the most celebrated moments of cognitive neuroscience in the past century. Giacomo Rizzolatti and his colleagues at the University of Parma, Italy, observed that some neurons in the premotor cortex (called F5) of macaque monkeys fired not only when the monkey performed a particular action, such as picking up a peanut with its fingers, but also when it watched another monkey or the experimenter performing the same action (Figure 5.8).⁵³ Observing the robust responses of these neurons leaves no doubt that they are special, and they represent a wide range of actions. In a small subset of mirror neurons, the correlation between the observed and executed actions is strong, both in terms of general action (e.g., grasping) and the executed action (e.g., precision grip). However, most neurons in the premotor area display a reliable correlation between the visual action they respond to and the motor response they organize.⁵⁴

53. The discovery of mirror neurons illustrates the importance of a prepared mind in research. There was no *a priori* hypothesis being tested since there was no way of knowing whether and where these neurons existed. The original observations were published by di Pellegrino et al. (1992), and the term “mirror neuron” was coined by Gallese et al. (1996). For a comprehensive overview of the mirror neuron system, see Rizzolatti and Craighero (2004). The existence of mirror neurons has also been used as a physiological support for the “theory of mind”; that is, the brain’s ability to detect other people’s (or animal’s) intentions, desires, and fears from their actions.

54. Disengagement of motor action from cortical activity can occur at a remarkable speed. In a brain–machine interface experiment, monkeys were trained to move a lever initially by hand and subsequently by producing neuronal activity in the premotor cortex, which was read out

Mirror neurons catapulted to fame after their indirect demonstration in the human brain. College students watched experimenters make finger movements and in some trials made the same finger movements themselves while their brains were monitored by fMRI. During both observation-execution (action) and observation-only conditions, activity was seen in two areas: the inferior frontal cortex (opercular region) and the rostral-most part of the right superior parietal lobe. Similar activations were observed when participants imagined movements instructed by symbolic cues. Although both brain areas participate in action imitation, their roles are somewhat different. The opercular region represents the observed action as a motor goal or plan, without defining the precise details of the movement. In contrast, the right parietal lobe represents the kinesthetic details of the movement (e.g., how much the finger should be lifted and in what direction). If these areas respond equally well to real and observed or imitated actions, how do people know if they are the actor or the imitator of a movement? One possible answer is that the signal in the parietal cortex reflects a corollary discharge copy of the planned action (Chapter 3), creating some firing pattern differences between real and internalized actions that can inform the brain that “it is my body that is moving.”⁵⁵

Extracellular recordings of neurons in human patients shed some light on such subtle differential patterns. A significant fraction of neurons in the supplementary motor area and even in the hippocampus respond while patients either execute or observe hand-grasping actions and facial emotional expressions. Some of these neurons were excited during action execution but inhibited during action observation. These findings are compatible with a mechanism that differentiates the motor and perceptual aspects of actions performed by self or others.⁵⁶

by a computer algorithm and sent to the actuator that controlled the lever. After a few hundred trials, the monkey's arm ceased to move and the lever was operated exclusively by neuronal activity (or, if you wish, the monkey's will; Nicolelis and Lebedev, 2009).

55. Iacoboni et al. (1999). These and other experiments on overt action and imagination are haunted by the possibility that the motor pathways and muscles are also activated in some attenuated form. The motor system is strongly connected to the rest of the brain, and, therefore, neuronal patterns underlying cognitions and emotions are often expressed in muscular activity even when it is inconsequential. Just watch how fans of football behave during an exciting match. They mimic the action of the players in a subliminal form. Before the study of Iacoboni et al. (1999), Rizzolatti and his colleagues examined hand muscle twitching and associated electromyograms while the participants watched the experimenter grasp objects. Often they found a good match between electromyogram patterns when the research participants watched the experimenter or actually grasped objects themselves (Fadiga et al., 1995).

56. Mukamel et al. (2010).

The original mirror neuron studies examined the similarity of neuronal patterns between performing actions and watching others perform actions. Subsequently, this idea spread to other areas of neuroscience, as researchers investigated how such neurons can encode other people's feelings, intentions, and emotions. When we see a tarantula crawling on someone else's chest, are the same type of neurons in the same brain regions active in both the observed person and the observer? fMRI experiments suggest so, at least in the secondary somatosensory area.⁵⁷ The mirror neuron system may link an action and its meaning or intent, whether the act is committed by yourself, by others, or even by simulation in the mind. The actions of these neurons may form the neurophysiological basis of some aspects of our social interactions.

The brain's disengagement from overt movement has important consequences. First, during internalized states, the motor system puts the action planning in a "motor format" (Chapter 11) so that it looks like a real action when related to recorded neuronal activity even when it is not accompanied by overt action. From this perspective, imitation can be regarded as a transformation of observed action into an internal simulation of that action. Second, the corollary discharge signals (Chapter 3) to parietal cortex, resulting in covert "actions," would allow the brain to evaluate the potential consequences of future plans. Without any overt behavior, the mirror and corollary discharge systems can then inform the brain's owner that she is the agent of this covert activity.⁵⁸

SOCIAL INTERACTIONS ARE ACTIONS

One of the most elaborate corollary discharge systems relates to eye movements (Chapter 3), which allow a newborn to begin to discover the world long before reaching and crawling develop. Faces and biological motion are especially effective in triggering eye movements in visual animals such as humans. There are at least two ideas to explain this. One view is that these cues are somehow

57. Keysers et al. (2004). Both feeling disgusted and watching an actor with a disgusted look activate the anterior insula (Wicker et al., 2003).

58. Dysfunction of the mirror neuron system has been suggested to underlie the derailed social interactions and symptoms of motor speech problems in autism (Iacoboni and Dapretto, 2006). The mirror neuron concept offers a potential framework for other dysfunctional states as well, such as misattribution of actions, hallucinations, delusion of agent influence, and experience of alien control in psychotic patients. The system of mirror neurons has been hypothesized to form a "core" network, including the primary temporal, posterior parietal, and prefrontal cortex, which becomes activated during action execution and simulated states, including observed and imagined actions (Wolpert et al., 1995; Jeannerod, 2001).

special for the brain, so they automatically trigger a response.⁵⁹ Another view is that eye movements reflect internally driven seeking for ethologically relevant patterns, so the brain can make adaptive use of those cues.⁶⁰ Although the distinction may seem subtle, the answers may determine how we search for and evaluate the underlying mechanisms. Monitoring eye movements in the laboratory tells us about the brain's engagement with the world. This simple method does not require verbal communication.

Alteration of eye movement patterns can be diagnostic for neuropsychiatric diseases, most prominently autism. Autism is usually defined not by its underlying mechanisms but by impaired social communication and repetitive behavior. One of the first things parents discover is that autistic children rarely solicit or reciprocate eye contact with caregivers, unlike normal infants. This difference becomes apparent by two to six months of age.⁶¹ There is nothing wrong with the eye movements in affected children; they simply look elsewhere.

There is no large gap between autism and many 'odd' but healthy people.⁶² Instead, there seems to be a continuum of sociality reflected in the amount of social visual engagement, which is largely determined by genetic factors. When identical twin toddlers viewed movies of other toddlers interacting, they tended to fixate their eyes on similar things. Moreover, they shifted their eyes at the same time in the same directions. In other words, their eye searches were remarkably concordant. Such joint patterns were much more rare in sex- and age-matched nonidentical twins or nonsiblings, showing that it is not only the movie scenes that synchronized the eye movement patterns in identical twins but also some internal programs. Heritability of eye search is most prominent in gaze toward the eyes and mouth. Toddlers with autism rarely focus on these critical parts of the face, so they do not learn about the social consequences of their actions from the smiles and grimaces of other children and caregivers who try to engage them.⁶³

59. This may be similar to the released action patterns in many animals (Tinbergen, 1951).

60. Scarr and McCartney (1983); Hopfinger et al. (2000); Treue (2003).

61. Jones and Klin (2013). In contrast to the active search idea, Markram et al. (2007) proposed that the amygdala is hyperreactive because people with autism find looking into the eyes of others to be stressful.

62. Some claim that scientists belong to the autistic end of the spectrum. Instead of making eye contact, scientists look at each others' shoes. Mathematicians, on the other hand, look at their own shoes.

63. Constantino et al. (2017).

Investigate the White of the Eyes

These findings, admittedly selected with some bias from an enormous literature with various and often conflicting views, suggest that internally driven seeking of face cues is heritable. The neuronal mechanisms responsible for social visual engagement are not well understood, but, if I may speculate a bit, the amygdala is likely to be a key structure. The amygdala and its partners comprise a neural system involved in the perception and expression of emotions⁶⁴ and the retrieval of social cues from facial expressions.⁶⁵ Nearly all social interactions involve emotions. Patients with complete bilateral amygdala damage have difficulty judging people's trustworthiness based on their appearance, compared to people with intact brains or control patients without amygdala damage. People with bilateral amygdala damage also cannot recognize fearful faces.⁶⁶

These results, along with numerous experiments in animals, have been interpreted to indicate that the amygdala is the seat of emotions, particularly fear. But this textbook teaching may be misleading. First, the impairment of emotional face judgments does not extend to verbal descriptions of people. Second, in contrast to the laboratory, where emotions are tested by viewing still faces, patients with amygdala damage do not necessarily misjudge untrustworthy persons in real life. Third, they easily recognize fear expressed via vocal communication. Fourth, when they are instructed to focus on the eyes in pictures, their judgment becomes close to that of control subjects.

64. Adolphs et al. (1994, 1998); Calder et al. (1966). Emotion is one of those constructs on William James's shortlist. This charged term has two aspects. One relates to actions and bodily expressions, which are perceptible and measurable by external observation; for example, breathing changes, flushing, crying, or other facial changes. The other aspect is an internal experience (or *qualium*), which is only available to the participant who has the emotion (Damasio, 1995). Only the observable aspects can be objectively studied (LeDoux, 2014). Brain imaging studies also point to the key role of amygdala in visual processing of emotional faces. Fearful faces induce positron emission tomography (PET) scan changes in the left amygdala, left pulvinar, left anterior insula, and bilateral anterior cingulate gyri (Morris et al., 1998). I cannot do justice here to the vast amount of thinking in this area of research, but I can recommend great summaries (Damasio, 1994; LeDoux, 2015). Classic sources on emotions are James (1890), Papez (1937), and MacLean (1970).

65. Perceptual models of autism suggest that the underlying problem is socioaffective oversensitivity.

66. Neuronal recordings from the amygdala in epileptic patients with comorbid autism show that the spike responses of neurons do not differ from those of a nonautistic group in general, nor does the responsiveness to whole faces. Yet some neurons in the autistic patients showed abnormal insensitivity to the eyes and abnormal sensitivity to the mouth (Rutishauser et al., 2013).

Both the clinical findings and the animal experiments are compatible with the alternative interpretation that the amygdala's main behavioral function is to direct the gaze to cues that are ethologically relevant for social communication. For primates, these cues are mainly aspects of the face (including the ears in monkeys). Physiological experiments support this idea.

There is a general consensus that the inputs that are physiologically relevant to neurons should induce robust firing. However, neurons in the amygdala are infamous for firing at extremely low rates under most circumstances, including fear conditioning experiments. Perhaps the experimental conditions that lead to fear learning are not really what amygdala neurons care most about. But similar to other neurons in the brain, amygdala neurons can also become very active under the right circumstances. In an illuminating experiment, neuronal activity from the monkey amygdala was recorded while monkeys viewed videos of natural behaviors in unfamiliar monkeys. When the recorded monkey made eye contact with the video monkey, the firing rates of many neurons increased several-fold. When the gaze of the movie monkey pointed instead to the peripheral parts of the recorded monkey's retina, only small responses were observed. A critical observation was that the firing latency of these gaze contact neurons varied between 80 and 140 ms, which is tens of milliseconds shorter than their response latencies to visual stimuli. This experiment therefore suggests that amygdala neurons do not respond to a particular visual pattern but may be involved in selecting which parts of the face should be consulted when examining strangers.⁶⁷ I interpret these findings to mean that emotions, which have traditionally been approached via sensory inputs, can be reevaluated when viewed from the action side of the brain. The disengagement hypothesis may be used as guidance here as well.

One of the oldest and most influential theories of emotion was put forward by William James and the Danish physiologist Carl Lange, known today as the James-Lange theory of emotion.⁶⁸ The theory argues that emotions occur as a result of physiological reactions to body signals that we normally do not detect, such as changes in heart rate and respiration. We run away not because we fear the bear but we fear because we run away from it. In William James's

67. Mosher et al. (2014). A dominant monkey initiates eye contact by staring at the eyes of others and waiting for them to return direct gaze. In contrast, submissive individuals avoid direct eye contact altogether or engage in it only briefly (Redican, 1975).

68. James (1884); Lange (1885/1912). Walter Cannon (1927) criticized the James-Lange theory, but he examined only sensory signals conveyed by the vagal nerve and spinal cord. More recent theories argue that emotions are cognitive interpretations, and the brain produces meaning from the autonomic responses in combination with prior experience and social cues (Barrett-Feldman, 2017; LeDoux, 2014, 2016; LeDoux and Daw, 2018).

words, “My thesis, on the contrary, is that the bodily changes follow directly the perception of the exciting fact, and that our feeling of the same changes as they occur is the emotion.” Not surprisingly, James’s explanation is “outside-in”; it is the body-derived signals whose conscious interpretation by the brain produces fear. The theory has been criticized multiple times over the past century, yet its appeal persists.

Emotion expert Joe Ledoux, working at New York University, recently discussed the distinction between brain mechanisms that detect and respond to threatening signals and those that give rise to conscious fear. Whereas the former can be studied in animals, such as pairing a place with an electric shock and examining neuronal responses during the process, fear is a subjective feeling in the conscious domain that cannot be pinned down by simply correlating neuronal activity with overt responses to those signals. I agree. However, we can attempt to draw a parallel between mechanisms of cognition and mechanisms of emotions. As we discussed earlier in this chapter, cognition is a disengagement of the brain from the motor action system. Similarly, emotion can be conceptualized as disengagement from the autonomous action system. In both cases, the fundamental physiological mechanism is the corollary discharge system; in case of emotion, it is the refference from the brain’s control mechanisms of visceral systems to the amygdala and neocortex. Although the phylogenetic roots of emotions may well be based on overt feedback from the autonomous system, more complex brains can disengage from it so that internalized feelings can occur in the absence of blushing, sweating, or pupil or respiration changes. In summary, internalization of the overt sensory feedback in the James-Lange theory can be an alternative way of conceptualizing and researching emotions.

LANGUAGE IS BASED ON INTERNALIZED ACTION

In social communication, individuals try to understand the gestural and vocal actions and intentions of others. If the mirror neuron system represents the actions and emotions of conspecifics, we can speculate that a similar framework may underlie the neurophysiological mechanisms of language, linking the gestural communication system of the limbs, ears, and face muscles to verbal communication. This may be the evolutionary reason that language is inherently linked with gestures. Look at anyone in a cell phone conversation or watching TV during a plane flight. There is no need for waving hands, smiling, or other gestures when the other party cannot reciprocate them. The brain just cannot help it and mobilizes the body anyway.

Providing anatomical support for the gestural communication hypothesis, area F5 in non-human primates and Broca's speech area in people are homologous brain regions. Broca's area may have evolved atop the mirror system for actions and provided a seed for generating and reading motor sequences of speech. Similarly, the parietal mirror system could be a precursor of the Wernicke's "perceptual" speech area. Through such evolutionary development, the combinatorially open repertoire of manual and body gestures could have provided scaffolding for the emergence of protospeech vocalizations, leading eventually to language with syntax and compositional semantics. The observation–execution matching system could have linked action to social communication, culminating in linguistic syntax, so that an effective sender-to-receiver relationship could develop between actor and observer.⁶⁹

In line with this hypothesized evolutionary path, action, as symbolized by the verb in language, is the architectural centerpiece of a sentence. You cannot master a language without learning the meaning of verbs. Verbs express intentions and encode what is happening in motion. As engines of syntactic action, they can have a large entourage in a sentence. A verb can specify the action or event, describe the trajectory of an action with respect to some reference spatial point, indicate whether an action is completed or ongoing, and select the semantic roles of the participants in an action. Verbs in many but not all languages have a tense, referring to past, present, or future. They express not only the intent but also the mood of the speaker. For example, ambulatory movement can be described by a variety of verbs that refer to different speeds, moods, intentions, directions, and patterns of walking—in essence allowing the listener to visualize the action.⁷⁰

The process that gives meaning to verbs and relates them to motor actions and real-world experiences is called "semantic grounding" (Chapter 3): an initially meaningless sound pattern acquires meaning to the listener. After this learning process, whenever the pattern recurs in the brain, it can bring back the simulated or constructed thing even when the thing (or episode) is not present.

69. Similar matching mechanisms could be involved in the recognition of phonetic gestures in humans (Gallese et al., 1996). This idea was further developed by Rizzolatti and Arbib (1998) and Arbib (2005) and has its antecedents in the "motor theory of speech perception" (Lieberman et al., 1967), which postulates the necessary access to the motor templates of words in their perception.

70. For example, walking, running, approaching, leaving, hopping, heading, trotting, trudging, passing, hitching, and fishtailing refer to different movement patterns, reflecting mood, speed, and direction. In action-oriented languages, such as Hungarian, more than 200 verbs describe the walk-run dimension.

The specific neural mechanisms underlying semantic grounding of nouns and words in general are not known but they are based on our interaction with the environment in unique ways. For example, “up” and “down” have positive and negative meanings, respectively, in most languages, likely because of a number of motor actions that lead to reward require an erect posture, combating the gravitation forces we live in. “She woke up”; “he fell asleep”; “I’m feeling down”; “his income is high, mine fell”; “things in the lab are going downhill”; or “she couldn’t rise above her emotions.” Numerous examples in language are based on action metaphors. “Digesting these pages” “he killed my brilliant idea”; “Einstein’s theory gave birth to a new era.”⁷¹

Early learning in human infants often takes place in the context of active exploration of objects. “Infants do not simply look passively at the jumble of toys on the floor but rather use their body—head, hands and eyes—to select and potentially visually isolate objects of interest, thereby reducing ambiguity at the sensory level.”⁷² Of course, the process of “reducing ambiguity” is the idea of action-assisted grounding or, as Wittgenstein noted it more eloquently: “Our talk gets its meaning from the rest of our activities.” Pointing by hand to objects or just gazing at them in humans is an efficient gesture to invite joint attention between communicating individuals.⁷³

Psychophysical, lesion, recording, and transcranial magnetic stimulation (TMS) studies suggest that motor cortex and somatosensory cortex are involved in linking action words to their direct meaning, whereas the inferior frontal cortex, temporal lobe, and inferior parietal cortex encode more abstract features of action words.⁷⁴

As in spoken language, auditory feedback when playing a musical instrument is necessary to learn how action leads to the desired effect. Copies of the planned action from premotor cortices are sent before the sound feedback arrives. Once this corollary system has been trained, sounds can be imagined

71. Lakoff and Johnson (1980).

72. Yu and Smith (2012). Pulvermüller (2003) also points out the connection between meaning and action. Reading the word “kick” activates the same part of motor cortex as an actual kick.

73. Pointing and gazing are examples of externalization of brain function in humans (Chapter 9). No such behaviors have been observed in other species, except in domesticated dogs. Miklósi et al. (2003) suggest that reading intention of the other is a result of human–dog co-evolution and is absent in wolves and even non-human primates. However, active communicative systems exist in all animals. Rats communicate the safety of food by breath. Once a “demonstrator” rat conveys information about a novel edible food, naïve “observer” rats exhibit substantial enhancement of their preference for that food even if they did not observe the demonstrator’s act of eating (Galef and Wigmore, 1983).

74. Damasio et al. (1996); Tomasino et al. (2008); Pulvermüller (2013).

even in their absence. This may explain why many deaf people can speak, sing, or even compose music if they lose their hearing after acquiring language, although they have difficulty regulating the volume and pitch of sound. Beethoven became virtually deaf by the age of forty-four, yet he continued to compose extraordinary music for the rest of his life.⁷⁵ In contrast, no composer born with hearing loss is known. The corollary feedback loop of the action system must be calibrated before it can be deployed in a useful way.

To recapitulate, the mirror neuron system allows us to read the intentions of others by interpreting body language. The expansion of the action system to spoken language permitted us to establish an extensive and effective communication system by interpreting speech, which can be viewed as a metaphoric form of action. The invention of language accelerated externalization of brain function, creating a collective species memory (Chapter 9).

To enable cognitive operations, the brain should be equipped with mechanisms to separate and combine neuronal messages, similar to the syntactical rules of language. How neuronal networks support these operations is the topic of the next chapter.

SUMMARY

We discussed several examples of how disengagement of brain networks from their external inputs can be useful for cognitive operations. The key physiological mechanism of this scenario is a corollary discharge-like system that allows the brain to interpret the activity of action circuits even in the absence of overt movement and sensory feedback from muscles. Within such an internalized world, brain networks can anticipate the consequences of imagined actions without the need to act them out. Instead, the outcomes can be tested against previously acquired knowledge, which creates new knowledge entirely through self-organized brain activity.

For the sake of simplicity, I portrayed externally and internally driven activities as separate processes that emerged sequentially during evolution. In reality, these steps are only abstractions. The logic of this framework is not that the networks of a small brain depend entirely on external inputs and larger brains switch to internal operations. Instead, the two processes are inherently intertwined, and the same circuit can perform both input-dependent and

75. Several other composers continued to create music after complete hearing loss. Bedrich Smetana created perhaps his most famous work, *The Moldau*, after his hearing had left him at age fifty.

input-disengaged operations.⁷⁶ For example, the head direction system can momentarily disconnect from environmental and body-derived cues to extrapolate the imagined head direction via internal mechanisms. Spatial navigation, especially dead reckoning, assumes some kind of memory. Larger brains may have more capacity for internal processing. Yet, even in a complex brain, memory recall can be improved by external information. When a narrative gets stuck, a simple cue comes in handy, which is why people prepare a list of keywords for talks. In other words, even small brains have elements of internal operations (“cognition”), but, as the complexity of neural networks increases, the share and efficacy of internalized computation also increases. Therefore, complex neuronal networks with numerous interactive loops in larger brains can predict the consequences of the brain’s actions over a much longer time scale and in more complex environments than smaller brains.

76. In fact, as discussed in Chapter 3, self-organized brain activity may be the fundamental brain operation that can generate useful motor outputs even without sensors (e.g., baby kicks). The modification of such self-organized patterns to external perturbations can provide “meaning” to the actions by matching preexisting neuronal patterns to action–perception (Chapter 13; Buzsáki, 2006). Experiments in developing ferrets show that the similarity between spontaneous and visually evoked activity in visual areas increases progressively with age, suggesting that *internal models* result from the adaptation of self-organized brain patterns to the statistics of the surrounding world (Fiser et al., 2004). In a similar vein, construction of an internal representation of the self as a special entity can be accomplished by reprocessing internal data simultaneously with information about the environment.

